# A valuation obstruction for $3 \times 3$ magic squares of squares, with a machine-checked proof

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**Abstract**

We study the open problem of whether a $3 \times 3$ magic square whose nine distinct entries are perfect squares exists. We reduce the problem to a purely additive condition on the set $D(e) = \{2xy : x^2 + y^2 = e^2\}$ attached to the center root $e$, and prove unconditionally that the center root of any such square must be divisible by at least two distinct primes $p \equiv 1 \pmod 4$: for centers $e = s\,p^a$ with a single such prime, the elements of $D(e)$ have pairwise distinct $p$-adic valuations, which is incompatible with the required additive structure. All results, including the reduction, are formally verified in Lean 4 with mathlib. We also report an exhaustive computation showing no magic square of squares has center entry below $10^{20}$, and that Bremner's fully magic square with seven square entries is, up to scaling, the only one with center root below $10^9$. Finally we describe a rigidity phenomenon that pins down the remaining two-prime case to explicit divisibility coincidences.

## 1 Introduction

Whether nine distinct perfect squares can form a fully magic $3 \times 3$ square is a well-known open problem, popularized by Martin Gardner's 1996 prize and studied extensively since; see Boyer's survey and problem collection [1]. The strongest results to date are computational (searches over magic sums and over arithmetic-progression parameters) together with structural constraints on hypothetical solutions due to Morgenstern, Roberts, Underwood and others [1].

Throughout, a *magic square of squares* is a $3 \times 3$ array of nine distinct perfect squares whose three rows, three columns and both diagonals share a common sum $S$. Summing the four lines through the center shows $S = 3E$ where $E = e^2$ is the center entry; we call $e$ the *center root*.

### Results

**Theorem 1** (Reduction)**.** *Let $D(e) = \{2xy : x^2 + y^2 = e^2,\ 0 < x < y\}$. A magic square of squares with center $e^2$ exists if and only if there are $u, v$ with $u,\ v,\ u + v,\ u - v \in D(e)$ (all nonzero, $u \neq v$); the square's entries are then $e^2,\ e^2 \pm u,\ e^2 \pm v,\ e^2 \pm (u + v),\ e^2 \pm (u - v)$.*

**Theorem 2** (Main theorem)**.** *If the center root $e$ is divisible by at most one prime $p \equiv 1$* (mod 4) *(to any power, and with arbitrary cofactor free of such primes), then no magic square* *of squares with center $e^2$ exists. Equivalently: the center root of any magic square of squares is divisible by at least two distinct primes $\equiv 1 \pmod 4$.*

**Theorem 3** (Computations)**.** *There is no magic square of squares with center entry $e^2 < 10^{20}$. Moreover, every fully magic $3 \times 3$ square with exactly seven square entries and center root $e < 10^9$ is an integer scaling of Bremner's example with $e = 425$.*

All statements of Theorems 1 and 2, including Fermat's theorem that three squares in arithmetic progression have common difference divisible by 24, are formally verified in Lean 4 over mathlib; the library (thirteen files, no unproven assumptions) accompanies this paper.

## 2 The reduction

Every $3 \times 3$ magic square has the form $c \pm \{0, u, v, u+v, u-v\}$ arranged as

$$\begin{pmatrix} c+u & c-u-v & c+v \\ c-u+v & c & c+u-v \\ c-v & c+u+v & c-u \end{pmatrix},$$

in which all eight line sums equal $3c$ identically. Thus magic is free and the entire problem is the squareness of the nine cells.

**Lemma 4.** *For $c = e^2$, both $c-d$ and $c+d$ are squares if and only if $d \in D(e) \cup \{0\}$ (up to sign). Indeed if $x^2 + y^2 = e^2$ then $c - 2xy = (x-y)^2$ and $c + 2xy = (x+y)^2$; conversely if $c - d = A^2$ and $c + d = B^2$ then $A^2 + B^2 = 2e^2$, $A \equiv B \pmod 2$, and $x = \frac{B-A}{2}$, $y = \frac{B+A}{2}$ satisfy $x^2 + y^2 = e^2$, $2xy = d$.*

Theorem 1 follows by applying Lemma 4 to the four center lines. (Distinctness of the nine entries corresponds to $u, v, u \pm v \neq 0$ and $u \neq v$.)

## 3 Proof of the main theorem

Fix a prime $p \equiv 1 \pmod 4$, write $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$ with $\pi = A + Bi$, $A^2 + B^2 = p$, and let $e = s\,p^a$ with $a \geq 0$ and $s$ free of primes $\equiv 1 \pmod 4$.

**Lemma 5** (Rigid part)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2$ is a unit multiple of $s$.*

*Proof.* Induct on the least prime factor $q$ of $s$. If $q \equiv 3 \pmod 4$ then $q$ is inert; $q \mid N(z) = z\bar{z}$ gives $q \mid z$, and we divide. If $q = 2$, then $1 + i$ divides $z$ twice (as $2 \mid N(z)$ twice), and $(1+i)^2 = 2i$ is 2 times a unit. □

**Lemma 6** (Classification)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2 p^n$ equals $u\,s\,\pi^j \bar{\pi}^{n-j}$ for a unit $u$ and some $0 \leq j \leq n$.*

*Proof.* Induct on $n$: for $n \geq 1$, $\pi \mid p \mid z\bar{z}$, so $\pi \mid z$ or $\bar{\pi} \mid z$; divide and recurse, finishing with Lemma 5. □

**Lemma 7** (Bridge)**.** *For every $m \geq 1$, $p \nmid \operatorname{Im}(\pi^m)$.*

*Proof.* Pick $\omega \in \mathbb{Z}/p$ with $\omega^2 = -1$ (possible as $p \equiv 1 \bmod 4$) and consider the ring maps $\varphi_\pm : \mathbb{Z}[i] \to \mathbb{Z}/p$, $i \mapsto \pm\omega$. If $p \mid \operatorname{Im}(\pi^m)$ then, since $\operatorname{Re}(\pi^m)^2 + \operatorname{Im}(\pi^m)^2 = p^m$, also $p \mid \operatorname{Re}(\pi^m)$, so $\varphi_\pm(\pi)^m = 0$; as $\mathbb{Z}/p$ is a field, $\varphi_\pm(\pi) = 0$, i.e. $A \pm B\omega \equiv 0$. Adding and subtracting gives $p \mid A$, $p \mid B$, contradicting $A^2 + B^2 = p$. $\square$

**Lemma 8** (Representation structure)**.** *Let $x^2 + y^2 = e^2 = s^2 p^{2a}$ with $xy \neq 0$. Then there are $1 \le t \le a$ and $\varepsilon \in \{\pm 1\}$ with*

$$2xy \;=\; \varepsilon\, p^{2(a-t)} \left(s^2 \operatorname{Im} \pi^{4t}\right), \qquad p \nmid s^2 \operatorname{Im} \pi^{4t}.$$

*In particular $\nu_p(2xy) = 2(a-t)$, and distinct elements of $D(e)$ have pairwise distinct $p$-adic valuations $\{0, 2, \ldots, 2a-2\}$.*

*Proof.* Write $z = x + yi$; by Lemma 6, $z = us\pi^j \bar{\pi}^{2a-j}$. Then $2xy = \operatorname{Im}(z^2)$ and $z^2 = u^2 s^2 \pi^{2j} \bar{\pi}^{2(2a-j)}$ with $u^2 = \pm 1$. If $j = a$ the product is real and $xy = 0$. Otherwise, with $t = |j - a|$, $\pi^{2j}\bar{\pi}^{2(2a-j)} = p^{2(a-t)} \cdot \pi^{\pm 4t}$, and $\operatorname{Im}(\bar{\pi}^{4t}) = -\operatorname{Im}(\pi^{4t})$; absorb signs into $\varepsilon$. Lemma 7 and $p \nmid s$ give the non-divisibility. $\square$

*Proof of Theorem 2.* Suppose $u, v, u+v, u-v \in D(e)$ as in Theorem 1, and give each the form of Lemma 8 with parameters $(t_i, \varepsilon_i)$ and constants $c_t = s^2 \operatorname{Im} \pi^{4t}$. If $t_1 = t_2$ then $u = \pm v$, excluded. Otherwise $\nu_p(u) \neq \nu_p(v)$; writing $u \pm v = p^{\min} \cdot (\text{bracket})$ where the bracket is $\varepsilon_i c_{t_i} + (\text{multiple of } p)$, the bracket is prime to $p$. Comparing with the exact forms of $u + v$ and $u - v$ and using uniqueness of exact $p$-power extraction, $2(a - t_3) = 2(a - t_4) = \min$, so $t_3 = t_4$ and $u + v = \pm(u - v)$, forcing $u = 0$ or $v = 0$. Contradiction. $\square$

*Remark* 9. The counting argument (four center lines require four representations) already forces two distinct primes *or* $p^4 \mid e$; the valuation argument closes the prime-power loophole entirely and is what we formalize.

## 4 Computations

Enumerating candidate centers directly through Theorem 1 is far cheaper than enumerating magic sums: factoring $e$ in windows and generating $D(e)$ from the Gaussian factorization checks all $e < 10^{10}$ (center entries to $10^{20}$) in minutes on a desktop; no configuration $u, v, u \pm v \in D(e)$ exists in that range. The same machinery classifies near-misses: configurations with two of the four differences in $D(e)$ and the other two "half in" (one square cell each) are exactly the fully magic squares with seven square entries, of which Bremner's $e = 425$ example was the only one known. Scanning all $e < 10^9$ finds 2,352,941 such configurations — every single one an integer scaling of $e = 425$, and none with eight square entries.

## 5 The minimal two-prime case

**Theorem 10.** *There is no magic square of squares whose center root is $s\,pq$, where $p \neq q$ are primes $\equiv 1 \pmod 4$, each to the first power, and $s$ is free of such primes.*

*Proof sketch.* Here $D(e)$ has exactly four elements: the interior conjugate pair $m, m' = s^2 |\operatorname{Im}(\pi^4 \chi^{\pm 4})|$ and the axis elements $X = s^2 q^2 |\operatorname{Im} \pi^4|$, $Y = s^2 p^2 |\operatorname{Im} \chi^4|$. Since $u, v, u+v, u-v$

are four distinct elements of $D(e)$, they exhaust it. Write $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$; recall $R_A, R_B$ are odd, $4 \mid I_A, I_B$, and by Lemma 7 $p \nmid I_A$, $q \nmid I_B$ (and $I_A, I_B \neq 0$). Enumerate the placements of $\{X, Y\}$ among $\{u, v, u+v, u-v\}$:

*(1)* $\{u, v\} = \{X, Y\}$. Then $\{u \pm v\} = \{m, m'\}$ and the conjugate-pair identities force $\{u, v\} = \{s^2|R_B I_A|,\, s^2|I_B R_A|\}$. Matching against $X, Y$ and multiplying the two resulting equations gives $|R_A R_B| = p^2 q^2$; but $|R_A| < p^2$ and $|R_B| < q^2$ strictly (as $I_A, I_B \neq 0$), a contradiction. The crossed matching forces $|R_A| = p^2$ outright, equally impossible.

*(2)* $\{u, v\} = \{m, m'\}$. Then $\{u \pm v\} = \{X, Y\}$ and $u+v,\ u-v \in \{2s^2|R_B I_A|,\, 2s^2|I_B R_A|\}$; matching yields either $2|R_B| = q^2$ or $4|R_A R_B| = p^2 q^2$, both impossible by parity.

*(3) mixed placements.* One axis element lies in $\{u, v\}$, the other in $\{u \pm v\}$. Each of the 64 sign/placement sub-cases yields two linear equations in $W = R_A I_B$, $Z = I_A R_B$; solving and substituting into the norm relations $(W/I_B)^2 + I_A^2 = p^4$, $(Z/I_A)^2 + I_B^2 = q^4$ leaves a difference that factors, in every sub-case, into terms from the list

$$(I_A q^2 \pm I_B p^2), \quad (2I_A q^2 \pm I_B p^2), \quad (I_A q^2 \pm 2I_B p^2), \quad 3I_A q^2, \quad 3I_B p^2.$$

The all-positive combinations cannot vanish; the difference combinations vanish only if $I_A q^2 \in \{I_B p^2,\, 2I_B p^2\}$ or $I_B p^2 = 2I_A q^2$, each of which equates a quantity prime to $p$ with one divisible by $p^2$ (Lemma 7). Hence no sub-case admits a solution. $\square$

*Remark* 11. Bremner's seven-square-entry example has center root $425 = 5^2 \cdot 17$, just outside the hypothesis (exponent 2): consistent with Theorem 10, the near-miss lives at the smallest center type the theorem does not reach.

## 6 Center roots $s\,p^2 q$: Bremner's type

**Theorem 12.** *There is no magic square of squares whose center root is $s\,p^2 q$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and $s$ is free of such primes.*

Here $|D(e)| = 7$: writing $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$ and $P = p^2$, $Q = q^2$, the elements are (up to the global factor $s^2$)

$$P\,|\mathrm{Im}(\pi^4 \chi^{\pm 4})|, \quad |\mathrm{Im}(\pi^8 \chi^{\pm 4})|, \quad PQ\,|I_A|, \quad Q\,|\mathrm{Im}\,\pi^8|, \quad P^2 |I_B|.$$

All seven are *linear* in $(R_B, I_B)$ over $\mathbb{Q}(R_A, I_A, P, Q)$. The four differences $u, v, u+v, u-v$ occupy four distinct elements; for each of the 840 placements and each resolution of the absolute values (6,720 linear systems in all), solving the two placement equations for $(R_B, I_B)$ and substituting into $R_B^2 + I_B^2 = Q^2$ leaves one polynomial relation in $(R_A, I_A, P, Q)$, which must vanish. A computer-assisted but fully rigorous sweep (scripts accompany this paper) shows every relation is impossible:

- 339 of the 343 distinct relation classes die by combinations of: parity ($R_A, R_B, P, Q$ odd, $8 \mid I_A, I_B$); the $p$-adic clash ($\nu_p$ of the two sides differ, using $p \nmid R_A I_A$ and Lemma 7); congruence obstructions modulo 16, 32 and small odd moduli on the curve $R_A^2 + I_A^2 = P^2$; the strict size pinch $|R_A|, |I_A| < P$; and elimination of $I_A$ via the norm followed by factorization. The finitely many exceptional primes ($\ell \in \{5, 13, 17\}$, arising from small constants in the eliminants) are closed by exact evaluation at the actual $\pi^4$.

- 4 classes reduce to homogeneous quintics/sextics in $(R_A, P)$; a rational root $R_A/P$ in lowest terms would need $P = p^2$ to divide a leading coefficient in $\{\pm 8, \pm 16, \pm 32, \pm 40\}$ — impossible.
- 4 classes are tautological (the norm residual vanishes identically); there the solved values are $(R_B, I_B) = \pm(Q/P)(R_A, -I_A)$, and integrality forces $p^2 \mid q^2$.

The 256 distinct determinants of the linear systems are proven nonzero by the same mechanisms, so no singular case escapes. □

*Remark* 13. Bremner's fully magic square with seven square entries has center root $425 = 5^2 \cdot 17$, of exactly this type: Theorem 12 shows the missing two entries can never be repaired at *any* center of that shape. Since the seven $D$-elements are linear in $(R_B, I_B)$ for every $b = 1$ exponent pattern $s\,p^a q$, the same pipeline applies to all $(a, 1)$ cases; the case count grows with $a$, and a uniform-in-$a$ argument is the natural next objective.

**Theorem 14.** *There is no magic square of squares whose center root is $s\,p^3 q$ ($p \neq q$ primes $\equiv 1 \bmod 4$, $s$ free of such primes).*

For $a = 3$ the difference set has ten elements, still linear in $(R_B, I_B)$; the sweep comprises 64,512 leaf systems in 3,767 relation classes. All 1,271 nonsingular classes die by the Theorem 12 mechanisms; the 2,496 singular classes (rank-deficient placements, which first appear at $a = 3$ through the three constant axis elements) die through their consistency relations by the same kills; and all 1,080 determinant classes are proven nonzero, so no singular locus escapes. The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ share one pipeline, and every two-prime exponent pattern $(a, b)$ is decidable in this sense — for $b \geq 2$ the leaf systems become zero-dimensional quadratic rather than linear, since $\chi^8, \chi^{12}, \ldots$ are polynomials in $(R_B, I_B)$. The open problems are uniformity in the exponents and the $\geq 3$-prime landscape, where the leaf systems acquire positive dimension.

# 7 The two-prime frontier

## Toward uniformity in the exponent

The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ suggest a uniform theorem for all $s\,p^a q$. Abstracting the levels of $D(e)$ into free circle variables linked by a chain map (the enriched relaxation), 95.8% of the 9,216 uniform two-level leaf systems die by mechanisms independent of $a$ — positivity pinches, parities, $p$-adic patterns and congruence obstructions. The residue is exactly 24 relations, independent of $a$, involving two $\pi$-powers $\pi^{4j}$, $\pi^{4(j+\Delta)}$. A complete weight-regime analysis shows that their phase-aligned leading forms cancel identically: the obstruction survives every first-order $p$-adic tool. The residue was then resolved further by the *exact orbit substitution*: under the two $p$-adic embeddings, $R_j - \omega I_j = \pi_-^{4j}$ exactly, so each relation becomes a finite Laurent polynomial with definite monomial valuations. Single-level templates die completely (unit leading coefficients; the tautological placements force $\operatorname{Im}\chi^4 = 0$; the $p = 5$ shapes die by exact 5-adic evaluation over a full period). Two-level templates die except on six explicit loci $\pi_+^{4\Delta} \equiv c \pmod p$, $c \in \{\pm 1, \pm 2, \pm\frac{1}{4}\}$; since $\pi_+^{4\Delta} = c$ exactly is impossible (norms), each locus kills all but finitely many $j$ per $(p, \Delta)$, and effective bounds on $v_p(\pi_+^{4\Delta} - c)$ — $p$-adic linear forms in logarithms — yield $j \ll \log^2 \Delta$: a complete effective-finiteness program for the uniform theorem over all $s\,p^a q$, with only the Baker-bound bookkeeping, the resulting finite verification, and the three- and four-level chain templates outstanding.

## Three and four levels: the $Q$-collapse and the leading pair

The three- and four-level chain templates are now closed to the same standard. A merged sweep (36,288 three-level and 31,104 four-level leaves) kills 95.7%, resp. 91.4%, by the zero-solution, imbalance, residual-unit and singular mechanisms. The residue collapses to 8, resp. 12, *deep shapes* — same-orientation interior elements plus one axis element — and for each of the twenty shapes a symbolic computation gives the decisive structural fact: writing the solved value as $\chi^4 = Q\,N/D$, the numerator has $Q$-degree exactly one and the denominator is $Q$-free. Setting $N = Q\,A$, the modulus condition $|\chi^4| = Q^2$ eliminates $q$ *entirely*: a deep-shape leaf survives if and only if

$$c \;=\; |A|^2 - |D|^2 \;=\; 0,$$

a single integer identity in the $\pi$-orbit algebra. Expanding $c$ over its monomial valuation groups $\gamma\,\pi^{P\cdot j}\bar{\pi}^{B\cdot j}$ (the substitution is exact, with no unit ambiguity), realness pairs the groups conjugately, and for *every* shape and *every* strong-separation ordering of the orbit indices the minimal-valuation set is exactly one conjugate pair with real coefficient $\gamma \in \{\pm 1, \pm 2\}$ — a unit for every odd $p$. The pair sums to $2\gamma\,p^{a_0}\,\mathrm{Re}(\bar{\pi}^{4M})$ and $v_p\big(\mathrm{Re}(\bar{\pi}^{4M})\big) = 0$, so $v_p(c)$ is finite: *every deep-shape leaf whose orbit indices avoid an explicit finite union of valuation-tie hyperplanes is impossible, uniformly in $p \geq 5$.*

On the tie set, layering $c$ by $\pi$-exponent reduces everything to integer *phase sums* $S_e(j) = \sum_i \gamma_i (2A)^{B_i\cdot j}$ (here $\pi = A + Bi$): survival requires $p \mid S_\pi(j)$ and $p \mid S_{\bar{\pi}}(j)$ simultaneously — a double multiplicative-order condition on $2A \bmod p$. A census over $p < 200$, orbit indices $\leq 6$, finds 1,806 such double-vanishing points among 362,880 (0.5%); exact evaluation shows $c \neq 0$ at every one, with even escape depths $v_p(c) - e_{\min} \in \{2, \ldots, 20\}$ governed by the multiplicative order of $2A$ — the special-value loci in exact arithmetic form. Independently, a direct sweep of the criterion over all twenty shapes ($p < 200$, indices $\leq 4$, $q < 500$; 3,236,352 exact checks) finds no survivor. The uniform program for $s\,p^a q$ thus stands closed at one and two levels unconditionally, and at three and four levels outside the thin double-vanishing set, where the small-order degeneration awaits its own (expectedly self-similar) leading-pair analysis.

## The tie-structure dichotomy

The hyperplane residue itself has now collapsed. Certifying the valuation-group geometry by exact rational linear programming (a phase-one simplex over $\mathbb{Q}$, no floating point), one finds for *every* one of the twenty deep shapes: every $\pi$-weight form carries a single group; no three forms can simultaneously achieve the minimum; and exactly *one* pair of forms is jointly-minimal-feasible — each shape owns a single tie hyperplane. On it, conjugation symmetry makes the minimal $\bar{\pi}$-layer sum to $(\gamma_1 + \gamma_2)\,t^e$ with $t = 2A \bmod p$. The certified coefficient pairs split the shapes in two:

- **Eight shapes have** $(\gamma_1, \gamma_2) = (1, 2)$: the layer is $3t^e \not\equiv 0 \pmod p$ for every $p \geq 5$, so $v_{\bar\pi}(c)$ is finite always and $c \neq 0$ — *these eight shapes are impossible unconditionally: all* $p \geq 5$*, all orbit indices, all* $q$*.*

- **Twelve shapes have** $(\gamma_1, \gamma_2) = (-1, 1)$: the layer vanishes identically, and the surviving locus is exactly the tie hyperplane, where the leading contribution is $\pi^e \bar\pi^{\,b}(\bar\pi^{\,\Delta b} - 1)$ and lifting-the-exponent gives the exact escape depth $v_\pi(\bar\pi^{\,d} - 1) + v_p(\Delta b/d)$ when $d = \mathrm{ord}_p(2A)$ divides $\Delta b$ (and death otherwise).

A census over $p < 200$, indices $\leq 6$ confirms the classification exactly: all 1,806 double-vanishing points lie in the twelve $(-1, 1)$-shapes, every one is the trivial cancellation (no mod-$p$ accidents anywhere), and $c \neq 0$ at each.

In fact the $(-1, 1)$ cancellation is *exact*: the tie $P{\cdot}j = B{\cdot}j$ forces both weight coordinates of the pair to coincide, so the two terms annihilate as Gaussian integers and $c$ equals the sum of the remaining groups on the locus. Running the leading analysis *recursively* — delete each exactly-cancelling pair, add its equality, and re-analyze on the sub-locus, terminating by dimension exhaustion (observed depth $\leq 2$) — closes every branch of every shape except finitely many explicit *order loci* (about nineteen in total across the twelve shapes): ties of opposite unit coefficients in a single coordinate with distinct exponents $b_1 \neq b_2$. There the layer sum is $\gamma(t^{b_1} - t^{b_2})$ with $t = 2A \bmod p$, zero precisely when $\mathrm{ord}_p(2A)$ divides $b_1 - b_2$, in which case the valuation deepens by exactly $v_\pi(\bar\pi^{\,b_1 - b_2} - 1)$ — a Fermat-quotient (Wieferich-type) quantity. Survival would require this deepening to outrun every remaining layer; the exact censuses refute it at every point in range. This is the complete structural characterization of the uniform $s\,p^a q$ frontier at three and four levels: finitely many Wieferich-type loci per shape, and everything else proven impossible.

## Both exponents two: the $(2, 2)$ program

For $e = s\,p^2 q^2$ the set $D(e)$ has exactly twelve elements with a clean two-dimensional grading,

$$D(e) \;=\; \big\{\, p^{4-2\alpha} q^{4-2\beta} s^2 \ \mathrm{Im}\big(\pi^{4\alpha}\chi^{\pm 4\beta}\big) : (\alpha, \beta) \in \{0,1,2\}^2 \setminus \{(0,0)\} \,\big\}.$$

Of the 95,040 placement-and-sign leaves, 93.5% die by the *two-prime grading lemma*: a relation whose minimal $p$-layer (or $q$-layer) holds a single term is impossible, since that term is a $p$-unit (modulo $\pi$, $\mathrm{Im}(\pi^{4\alpha}\chi^{4\beta})$ reduces to a unit multiple of $\bar\pi^{4\alpha}\bar\chi^{4\beta}$) standing against terms divisible by two more powers of $p$. The 6,208 residual leaves collapse to 216 distinct three-term relations, and an exact sweep over every ordered pair of primes $p \neq q \equiv 1 \pmod 4$ below 1000 (39,234,560 checks) finds none satisfiable: *no magic square of squares has center root* $s\,p^2 q^2$ *with* $p, q < 1000$*, for any* $s$*.* The uniform closure of the 216 relations reduces to coupled residue conditions on $(\chi \bmod \pi,\ \pi \bmod \chi)$ — the same Wieferich-flavored endpoint as the deep loci above.

The grading lemma is exponent-independent, and its reach *grows* with the exponents: for general $e = s\,p^a q^b$ the set $D(e)$ is always $\{p^{2a-2\alpha} q^{2b-2\beta} s^2 \ \mathrm{Im}(\pi^{4\alpha}\chi^{\pm 4\beta})\}$ of size $((2a{+}1)(2b{+}1) - 1)/2$, and the census of leaves killed by the lemma reads 93.5% at $(2, 2)$, 96.2% at $(3, 2)$, 97.7% at $(3, 3)$, and 98.9% at $(4, 4)$, with the residual relation count growing only polynomially (216, 468, 1008, 3040). Thus the *entire* two-prime landscape reduces, uniformly in the exponents, to explicit doubly-balanced relation families carrying coupled-residue survival conditions.

For $e = s\,p^a q^b$ with two distinct primes $\equiv 1 \pmod 4$, the valuation pairs $(\nu_p, \nu_q)$ organize $D(e)$ into *conjugate pairs* $\{d_{j,k}, d_{j,-k}\}$ of multiplicity exactly two off the axes, and axis singletons. If $\{u+v, u-v\}$ is a conjugate pair $\{d, d'\}$ arising from $A = \pi^{4j}$, $B = \chi^{4k}$, then

$$d + d' = 2\,\mathrm{scale} \cdot |\operatorname{Re} B \ \operatorname{Im} A|, \qquad d - d' = 2\,\mathrm{scale} \cdot |\operatorname{Im} B \ \operatorname{Re} A|,$$

so $u$ and $v$ are *completely determined* — and their valuation pair coincides with that of $\{d, d'\}$, forcing $u, v \in \{d, d'\}$, which is absurd. Consequently every hypothetical two-prime solution requires an explicit divisibility coincidence ($q^t \mid \operatorname{Re} \pi^{4j}$, $p^t \mid \operatorname{Re} \chi^{4k}$, or a mod-$p$ or mod-$q$ cancellation in $u \pm v$). These coincidences recur along arithmetic progressions of exponents, so closing them requires exact-magnitude input beyond valuations; we leave this as the principal open direction, noting that together with Theorem 2 it would force every solution's center root into three or more primes $\equiv 1 \pmod 4$.

## References


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[3] M. Gardner, *The magic of 3x3*, Quantum 6 (1996), 24–26.